

Understanding Risks of Data Science Failures through Sociotechnical Approach

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ABSTRACT

A long-standing question for CHI and CSCW community is to understand data science work and everyday practices. Based on six months of in-person and virtual ethnography with a private organization located in United States and India, this study explores how data science practitioners articulate and manage risks of data science project failure. Aligning with the plurality of risk articulations and power laden risk protocols, the research uses sociotechnical lens to explore the affordances between social actors and technology for risk management. The multiple affordances identified will inform strategies to help practitioners in everyday risk identification and management. The goal of this research is to offer empirical insights and actionable frameworks that will assist practitioners in navigating complex sociotechnical risk management landscapes.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing; Empirical studies in collaborative and social computing.

KEYWORDS

Data Science Work, Risk Management, Multi-sited Ethnography, Global South

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1 BACKGROUND AND MOTIVATION

The HCI and CSCW research communities have placed significant emphasis on understanding work practices within a data science project [1]. The exploration has become more prominent amid growing challenges related to data science project harms and failures. This research probes the overarching question of how social actors from Global South and Global North who engage in heterogeneous data science work practices ascribe and manage the risk of data science project failure. The focus is not only on how the attribution of risk of data science failure is aligned with colonized

and regulated definitions of risk, but also on the deviation and resistance in a particular socio-spatio-temporal setting. As the risk of data science failure can be viewed in multiple dimensions, the determination of risk rests on what a practitioner anticipates as risk and how they assess the materialization.

Being a control measure, risk, power, justice, and legitimacy are interconnected and consequently, risk is politicized in the context of accountability (or responsibility), blame [32] and consensus [28]. Legitimation of risk is an exercise of power that normalizes practitioners to act around risk mitigation strategies [2]. The present study tries to unravel the risk performativity by understanding the articulation of risk on practitioner's level, and how is risk managed through data science work practices [3].

To summarize, some of the questions that will be addressed through this research are:

- how do practitioners make the 'discretionary judgements and negotiations' [4] to legitimize risk of failure in everyday data science work,
- how is the intersection of gender, education, subject matter expertise and location related with the legitimization of risk, could the intersection of identities privilege one and discriminate the other while confirming risk,
- how is the risk of failure managed on individual and institutional level

Following the interpretation of risk, the goal of this research is to finally understand how sociotechnical affordances can help create actionable frameworks to manage risks of data science project failure.

2 RELATED WORK AND RESEARCH QUESTIONS

2.1 IS Project Failure

There is a substantial body of literature accumulated over forty years on Information System (IS) failures. The IS scholarship broadly constitutes three areas of determinants that contribute to IS failure: people, organization, and technology; with processes underlying across all three [5]. At a macro level, technocentric failure can be considered as a key driver that leads to data science project failure [6]. However, the lens of technological determinism draws heavy criticism from scholars in sociology of scientific knowledge (SSK) and science and technology studies (STS) for its neglect of social factors and human agency [7, 8]. Although the unprecedented development of data science creates efficiency and automation of human work to a larger extent, STS scholars argue that technological development is not a linear process where the data science systems could be studied standalone. A sociotechnical

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approach must be taken to analyze the human agency in the making of the systems, and how the systems in-turn mediate the actions of actors.

2.2 Data Science Project Failure

One key driver, not typically encountered in other IS projects, are the data science projects that are labeled as ‘successful’ and yet deploy discriminatory models. These models can perpetuate ethical, reputational, regulatory (e.g., GDPR), legal, privacy and social risks [9]. In other words, there are multiple definitions of data science failures standardized by a diverse set of actors and institutions.

Needless to say, failure of data science projects is a non-linear concept which can be defined in different contexts. From the outset, data science systems as BlackBox or ‘[opaque box]’ [10] are difficult to decode into smaller digestible constituents to assess the failure. Also, because the codes are proprietary, it is challenging to get through the logic behind the creation of algorithms for data science projects to analyze failure. Failure is an outcome that cannot be retracted once realized. It is therefore necessary to understand the driving forces that may create a likelihood of failure. In other words, it is more important to concentrate on the anticipated risk of failure than the failure itself as an outcome.

While the repercussions of data science project failures can have detrimental consequences to the user base, the scope of this study concentrates on the root causes of risks of failure that emerge right at inception and during the project workflow with practitioner’s engagement.

2.3 Data Science Project Risk of Failure Identification

One of the ways to explore the risk of failure could be to investigate (1) who is behind the creation, who is not and (2) how data science practitioners work through the lifecycle to create the systems [10]. Data science projects typically follow streamlined or ad-hoc workflows of human led practices [11–13]. The resultant of working through the task-based phases of - preparation, modeling, and deployment phase [12] - with risk negotiation ultimately leads to creating data science systems. Various machine learning and statistical tools are used in the lifecycle to devise the system. In the workflow, the data science practitioners engage in understanding the business problem to determining the right model for devising the system. The practitioners also termed as ‘omnivore identities’ [14] can juggle between several roles, each with their unique social capital [15]. These roles can include subject matter experts, data scientists, data analysts, software engineers, project managers, or business analysts depending upon their engagement on the project [10]. Throughout the process of building the data science systems, there are ‘discretionary judgements and negotiations’ [4, 16] made by the practitioners to (de)legitimize risk of failure. The distinct sense-making give rise to practitioner’s individual articulation of risk which may or may not be different from the established codes or protocols of risk in the institution. Also, the risk of failure can emerge at any phase of the lifecycle. It is important to acknowledge that various risks may also be known to social actors in advance and considered taken-for-granted or

benign in the workflow. The reasons for tolerating the risk during the work cycle can be in lieu of gaining prospective rewards or cost effectiveness.

Similar to data science work practices being non-linear, the risk of failure is multidimensional. Risk may be subjected to the discretion of social actors who situate themselves at various levels of power hierarchy and contribute to decision making process. The risk of data science failure is spatial and temporal that is not restricted to time and space. Risk becomes more intricate to understand when the process of legitimization is transnational in nature. For example, the ‘data work’ prevalent in Global North becomes very different than what is prevalent in Global South [17]. The differences of discourses and power inequality with respect to transnational politics may influence how the risk is articulated and confirmed in local and global structures.

2.4 Previous Research on Data Science Project Focused on Risk Articulation

Drawing on observations from semi-structured interviews in conducted in 2020 with 16 male US data science practitioners with a partial view on risk [18], this research builds on CSCW and CHI [33] scholarship to understand the precursor of data science project failure which is the risk of failure. The conceptualization and the ontological stance of risk is heavily contested for the blurry boundary of risk being real or risk being constructed. Using the lens of science and technology studies, sociotechnical theory, sociology of work, sociology of culture, feminist theory and critical data studies, this multi-sited immersive ethnography in India and US captures the interconnectedness and messiness of social relationships and data science systems that shapes risk articulation and risk management on individual and institutional level.

The foundation of this research was laid out in 2020 with a qualitative interview study with 16 male data science practitioners [18–21]. The objective of this study was to explore (1) the existing risk management processes for data science projects established in public and private sectors for data science practitioners (2) how risk was articulated by data science practitioners. Following the inductive approach, it was observed that risk was articulated differently by different practitioners depending upon their team profiles, length of work experience in data science and risk management, and subject matter expertise. It was also observed that risk was not necessarily always used as a formal dialectical term. The practitioners conducted risk management strategies without terming the practices as risk management. For example, one of the participants did not feel comfortable to even using the word ‘risk’ [18].

A limitation of this study was the lack of diversity in the participant pool. As we could not find an availability of women data science practitioners for this study, the semi-structured interviews were conducted with only male data science practitioners. As a result, the themes were skewed towards a specific gender in the study. In the upcoming extended research grounded in corporate ethnography [22], the study will aim at answering following three overarching questions: (1) how is the risk of data science project failure articulated and managed on an individual level? (2) how is risk of failure articulated and managed on institutional level? (3)

what sociotechnical affordances can be used to effectively manage the risks of data science projects?

The fieldwork will be conducted in a private US based cloud computing consulting organization. There are two locations selected for the field work – India and the West Coast – to determine how data science practitioners articulate and manage transnational risk on individual and institutional level. The approach will be participant observation through which I will explore how during the project execution, risk is legitimized and delegitimized among data science practitioners. The focus will also be on determining if gender has any contribution towards the negotiation of risk on a team level.

3 RESEARCH APPROACH, METHODS, AND RATIONALE

There is a heightened awareness by HCI researchers for improving the efficiency of data science project workflow with human-centered design methods. AutoAI or AutoML [12] for automating data science practices, AutoDS to automate ML techniques [11], MAGNETON [23] to bring transparency to data science workflows are some of the design layers added to streamline and expedite data science work. While there is a focus on efficiency and automation, how the making of disruptive algorithms during the lifecycle opens likelihood of risk of failure on institutional and individual level still lacks extensive research. My research looks into how risk is anticipated as the project progresses from data understanding to deployment, and how risk is interpreted by different practitioners who get involved in the project.

Critical data studies scholars partially address risks of data science projects with challenges of trust and credibility [24], code hoarding, code cloning and task switching with IDE [25], and data quality while annotating ground truth [26]. My research on the other hand looks at the entire lifecycle of the project and explores how data science practitioners articulate risks that may or not align with the institutional risk management policies and toolkits. As shown with the themes in 2020 qualitative study [18] that highlights different risk articulations, I extend the study with the fieldwork to analyze if there is a deviation in risk articulations from standard definitions and, if the risk management strategies followed by practitioners is different than what is already established by the institution.

The research methodology of the fieldwork comprises of two dimensions: (1) studying distinct data science projects and (2) examining the perspectives of social agents by locations through field studies and interviews [27]. The study consists of six months of multi-sited corporate ethnography fieldwork both in person and virtual settings in India and the United States. The practitioners will be interviewed and be asked to walk me through the tools used for data science projects. Various questions on conflict on risk legitimization and optimal strategies on risk management will be asked upon participant's consent.

The fieldwork is scheduled to be conducted in a US-based cloud computing and technology services provider organization. This private sector consulting organization is based in the United States and maintains subsidiaries in more than 20 locations across the globe.

3.1 RQ1: Risk Articulation and Management on Individual Level

As the definition of risk is regimented with regulations and policies, the practitioners tend to get influenced by the normative risk governance. [27] uses the term 'regime' or 'regimes of truth' that shape individual's thoughts and actions in a specific way. Drawing on Foucault's governmentality or systems of power that develop 'regimes of truth' [27, 28] of risk through regulations and institutional protocols, the first research question tries to critically analyze how data science practitioners form their opinion on risk. The study also explores if the practitioners create any resistance against normative risks by crafting their own strategies or hacks for risk mitigation in everyday practices. Through the fieldwork, I aim at exploring how data science practitioners act upon what they construe as risk and how they organize data science work to manage the risk of failure for the project.

The evidence of different individual articulation is witnessed in the qualitative study done in 2020 with 16 male data science practitioners located in United States. The practitioners articulated risk through six themes [20] - Ethical Risk, BlackBox Models, Data Privacy, Data Storage, Financial Risks, and Success Criteria. Although there were established guidelines and policies in place for risk, the practitioners interpreted risk based on their expertise, responsibilities, and nature of the organization (sector and industry).

In the upcoming study, the aforementioned themes would be triangulated with corporate fieldwork and interviews will be conducted with 50 to 70 participants to determine how practitioners define risks of failure for data science projects in India and United States. As the practitioners carry with them multitude of skills, working experience and social capital [15], this study also explores how the agency of the practitioners with their gender, social capital, and location play a part with risk negotiation and legitimization.

3.2 RQ2: Risk Articulation and Management on Institutional Level

According to Beck, we live in a risk society [29] where individuals are aware of risk and always confronted with fear of hazards. Various policies, toolkits, and regulations are created to combat the fear of failure. Likewise, risk has a normative definition that is established in the institutions to be adhered to and managed by data science practitioners. For example, in the qualitative study done with 16 male data science practitioners, one of the participants from finance industry [18] construed Federal Reserve's Matters Requiring Attention (MRA) as the risk that had to be adhered by the team to manage. In my research, I aim at understanding how risk is defined on an institutional level, and how risk management strategies are skewed towards the established policies and protocols. The goal of crafting this research question for my study is to understand how reification of data science risk becomes an organizing principle for data science risk management strategies.

3.3 RQ3: Sociotechnical Affordance for Risk Management

Gibson [30] construed the concept of affordance which creates a relational property between the social agent and the technology.

An affordance between the data science practitioner and the object (e.g., Jupyter Notebook) refers to the attributes of both Jupyter Notebook and the data science practitioner. In other words, Jupyter Notebook affords the actions of writing codes and previewing to check after the code is functional. The interaction between the Jupyter Notebook and the data scientist determines its affordances [31]. Sociotechnical systems afford the relational property between the technology and the social actors. The affordances here develop the potential for action and contextual relationship between the actor and environment system. Under the affordance perspective for Jupyter Notebook, the question is not restricted to how Jupyter Notebook hosts the needs of all data scientists, but how Jupyter Notebook is a product of many affordances that enhance the features of technological artifacts. In my research, I look at the relationship between technology and the social agents to determine how the latter affords the technology, and how those affordances could be used to manage the risks of data science project failure.

4 RESULTS AND CONTRIBUTIONS TO DATE

4.1 Study 1: Lack of Risk Management Framework for Data Science Projects

The foundation of this study was laid out in 2020 with systematic literature review (SLR) to explore the presence of risk management framework for data science project [19]. The search was performed over six repositories ACM digital library, Google Scholar, Science Direct, Scopus, Web of Science, and IEEE Xplore published within a span of five years from 2015 to 2020. Upon search, the themes gathered through content analysis did not highlight any strong theme of a dedicated risk management framework for data science project. As this research was incomplete without getting the perspectives of social agents who were involved in creating data science systems in the real-time set-up, a second follow-up interview study was pursued in 2020 with data science practitioners [18].

4.2 Study 2: Data Science Risk Management Process existing in Private and Public Organizations

An empirical study was conducted in 2020 with 16 male data science practitioners across both public and private organizations to determine how DS project failure risks were managed [18]. Consistent with the SLR, inductive themes highlighted that no dedicated risk management framework was used for managing data science project failure risk. Yet, project methodologies in different capacities were used to manage the DS risk. Interestingly, a key theme that emerged during inductive analysis was that there were multiple interpretations of failure risk for DS projects. For example, whereas in defense agency, the enterprise risk level assessment was formulaic and managed via project management, a data science manager in a private finance organization defined risk as ad hoc and risk was managed with need based informal meetings. The presence of the risk management process was also dependent upon the maturity of the data science team within the firm and the subject matter expertise in risk management domain. In short, the analysis highlighted different articulations of risk from professionals working across different sectors and industries.

5 EXPECTED NEXT STEPS, OPEN QUESTIONS AND CHALLENGES

There are three research questions of this study out of two that are based on the themes I gathered during the qualitative study in 2020 [18]. As a next step, I plan to explore how the third research question on sociotechnical affordances can be addressed with the technology data scientists are using for their project. By the time of CHI 2024 consortium, the fieldwork in India and semi-structured interview with data scientists will be complete. The next step will be virtual ethnography conducted with data scientists in management teams located on the West Coast. Based in United States as a Global South scholar and aspiring to conduct fieldwork in India, I am currently facing the identity crisis of situating myself as emic-epic identity.

6 DISSERTATION STATUS AND LONG-TERM GOALS

I am currently enrolled as a doctoral candidate pursuing my fifth year of PhD in School of Information Studies in Syracuse University. My advisor is Professor Jeff Saltz. The doctoral program is a five-year commitment, and I am delaying my graduation for a year. I plan to graduate by Spring 2025.

My long-term goal is to conduct research in human-centered data science and ethical data science from the risk management perspective. After my graduation, I am interested in joining academia as an assistant professor in one of the schools.

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